

Yeu Wen Mak
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Dear Hiring Manager,

I am writing to express my strong interest in the Research Internship Program at ASPI, commencing January/February 2026, as advertised. My academic and professional trajectory has converged on analysing how open-source intelligence, data systems and complex socio-technical interactions shape policy outcomes and strategic decision-making. ASPI's emphasis on publicly accessible research, forward-looking strategic insights, and open-source modelling aligns directly with my experience and intellectual orientation.

Over the past year I have served as a Graduate Data Scientist with a Commonwealth government agency, completing three rotations that developed rigorous analytical and methodological frameworks. In Rotations 1 and 2 I designed a Minimum Description Length (MDL)-based thematic analysis pipeline to extract structured stakeholder arguments from consultation responses to the Privacy Act Review. The approach compressed qualitative submissions into quantifiable themes, enabling the mapping of argument clusters (e.g., "re-identifying de-identified information", "consent for precise geolocation tracking") with frequency, exemplar quotes and semantic identifiers. This work demonstrates my ability to design reproducible, transparent methods for interpreting public discourse—a capability directly transferable to open-source intelligence, trend-spotting and policy analysis.

In Rotation 3 I built an end-to-end methodology for analysing compliance behaviour under the Modern Slavery Act, integrating multiple datasets (ABR, ASIC, ATO, register logs) into four foundational universes—Identity, Obligation, Action, Governance. The output: a one-row-per-entity "master behavioural file" that enabled profiling of compliance status, governance risk and behavioural patterns. That task demanded not only technical data-engineering and modelling but also a systemic view of how legislation, governance networks and corporate behaviour adapt in response to regulatory regimes—precisely the sort of system-level insight ASPI values.

My earlier academic background—a Master of Science in Financial Engineering—equips me with applied quantitative and econometric skills, while training at the Santa Fe Institute introduced me to viewing the economy and institutions as adaptive complex systems. I framed entities, regulations and markets in terms of feedback loops, emergent behaviour and regime shifts. This "Quantum-Complexity-Entropy-Adaptation" (QCEA) paradigm underpins my analytical lens: I look for how information flows, how system states change over time, how adaptation occurs, and how open-source signals can reveal early-stage transformations in strategic landscapes.

At ASPI, I seek to apply these competencies by contributing to research projects that monitor emerging technology-geopolitics, track capability development, map influence vectors, and surface early signals of systemic change. I am motivated by ASPI's open-source approach—no security clearance required, enabling transparency, public accountability and wide dissemination of findings. My value proposition to ASPI includes:

- Designing and implementing analytic pipelines that convert raw open-source data into structured intelligence;
- Applying systemic modelling frameworks to capture feedback loops between technology, policy, regulation and strategic behaviour;
- Communicating complex findings in clear, policy-relevant language and visual forms;
- Collaborating across interdisciplinary teams, synthesising technical, legal, strategic and policy domains.

The internship at ASPI will allow me to deepen my skills in open-source strategic research, contribute to high-impact public work and engage with cutting-edge questions about technology, strategy and policy. In return, my home agency will benefit when I bring back capability in transparent data-driven modelling, stakeholder discourse interpretation, system-level analytical frameworks and open-source intelligence mindsets. Ultimately, I seek to bridge government analytical architectures with emergent public research modalities where innovation, transparency and strategic insight co-exist.

Thank you for your consideration. I have attached my résumé, academic transcript and the requested referee details. I look forward to the possibility of contributing to ASPI's research community.

Yours sincerely,
Yeu Wen Mak